FAULHABER

38 mNm
20 W

Brushless DC-Flat Motors

External rotor technology, with housing



Series 3216 ... BXT H

Values at 22°C and nominal voltage	3216 W		009 BXT H	012 BXT H	024 BXT H	
Nominal voltage	U_N		9	12	24	V
Terminal resistance, phase-phase	R		0,55	0,88	3,26	Ω
Efficiency, max.	$\eta_{max.}$		83	84	81	%
No-load speed	n_0		6 060	6 230	6 250	min^{-1}
No-load current, typ. (with shaft ø 4 mm)	I_0		0,165	0,126	0,068	A
Starting torque	M_A		225	245	263	mNm
Speed constant	k_n		691	530	267	min^{-1}/V
Back-EMF constant	k_E		1,45	1,89	3,75	mV/min^{-1}
Torque constant	k_M		13,8	18	35,8	mNm/A
Current constant	k_I		0,0724	0,0555	0,0279	A/mNm
Slope of n-M curve	$\Delta n/\Delta M$		27,5	25,9	24,3	min^{-1}/mNm
Terminal inductance, phase-phase	L		191	331	1 290	µH
Mechanical time constant	τ_m		5,28	4,97	4,66	ms
Rotor inertia	J		18,3	18,3	18,3	gcm²
Angular acceleration	$\alpha_{max.}$		123	134	144	·10³rad/s²
Operating temperature range:						
– motor		-40 ... +100				°C
– winding, max. permissible		+125				°C
Shaft bearings		ball bearings, preloaded				
Shaft load max.:						
– with shaft diameter		4				mm
– radial at 3 000 min^{-1} (5 mm from mounting flange)		15				N
– axial at 3 000 min^{-1} (push / pull)		3				N
– axial at standstill (push / pull)		50				N
Shaft play:						
– radial	≤	0,015				mm
– axial	=	0				mm
Mass		65,3				g
Direction of rotation		electronically reversible				
Speed up to	$n_{max.}$	10 000				min^{-1}
Number of pole pairs		7				
Hall sensors		digital				
Magnet material		NdFeB				

Rated values for continuous operation						
Rated torque	M_N		37	38	38	mNm
Rated current (thermal limit)	I_N		2,76	2,18	1,1	A
Rated speed	n_N		3 400	3 860	4 320	min^{-1}
Rated slope of n-M curve	$\Delta n/\Delta M$		71,9	62,4	50,8	min^{-1}/mNm

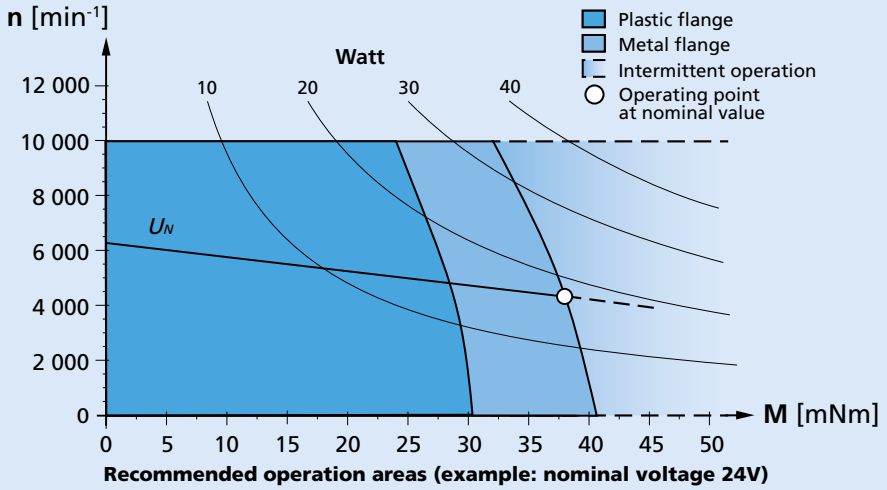
Note: Rated values are measured at nominal voltage and 22°C ambient temperature.

Note:

The display shows the range of possible operation points of the drives at a given ambient temperature of 22°C.

The diagram indicates the recommended speed in relation to the available torque at the output shaft.
It includes the assembly on a plastic- as well as on a metal flange (assembly method: IM B 5).

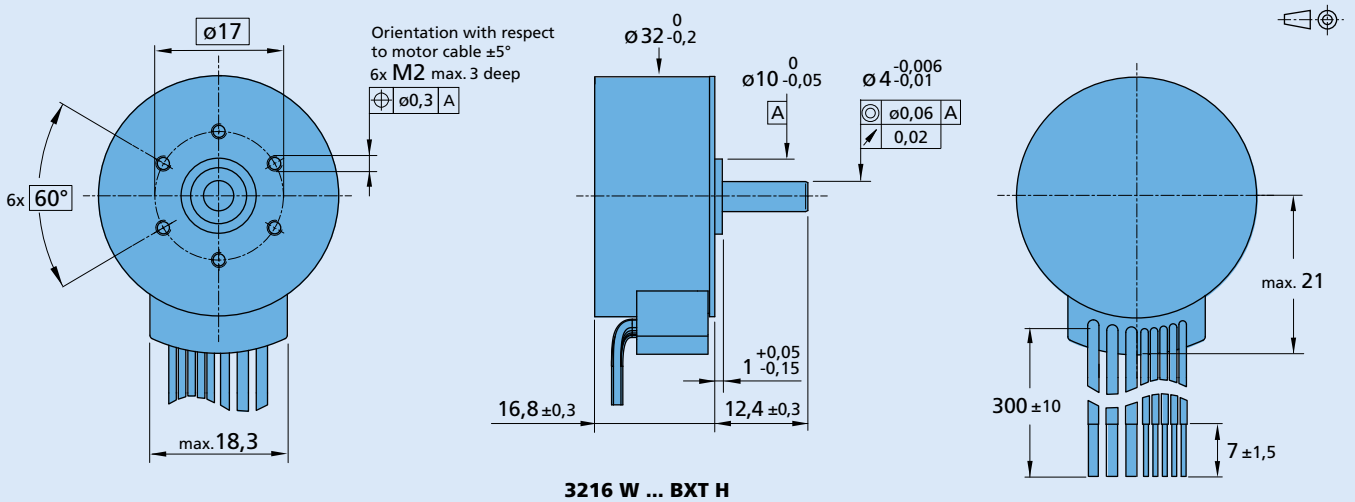
The nominal voltage linear slope describes the maximal achievable operating points at nominal voltage.
Any points of operation above this linear slope will require a supply voltage $U_{mot} > U_N$.



Recommended operation areas (example: nominal voltage 24V)

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Dimensional drawing



3216 W ... BXT H

Option, cable and connection information

Example product designation: **3216W012BXTH-3830**

Option	Type	Description
3830	Connector 	Standard cable with connector MOLEX Microfit 3.0, 43025-0800, recommended mating connector 43020-0800

Connection

No.	Function	Colour
1	Phase C	yellow
2	Phase B	orange
3	Phase A	brown
4	GND	black
5	U_{DD} (+5V)	red
6	Hall sensor C	grey
7	Hall sensor B	blue
8	Hall sensor A	green

Standard cable

Single wires, material PVC,
AWG 20, Phase A/B/C
AWG 26, Hall A/B/C, U_{DD}, GND

Product combination

Precision Gearheads / Lead Screws	Encoders	Drive Electronics	Cables / Accessories
22GPT 22GPT LN 22GPT HT 26A 26/1R 32GPT 32GPT LN 32GPT HT 32/3R 22L ... ML 22L ... SB 22L ... PB 32L ... TL 32L ... ML 32L ... SB 32L ... PB	IE3-1024 IE3-1024 L IEF3-4096 IEF3-4096 L IERS3-500 IERS3-500 L IER3-10000 IER3-10000 L IERF3-16384 L	SC 2402 P SC 2804 S MC 3602 B MC 3603 S MC 3606 B MC 5004 P MC 5005 S	PMB32 To view our large range of accessory parts, please refer to the “Accessories” chapter.

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